

Relevance-Based Prediction: Factor Selection for Saugus Schools

Cross-section year: FY2024 — the latest year with complete MCAS outcomes and DESE per-pupil spending data

Report run: July 9, 2026 · Final year of data utilized: FY2024

- Four outcomes: MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, Education Budget Share
- One RBP run per outcome over the full candidate pool— no in-model pruning (faithful to Kritzman)
- All curated factors enter one tier-blind RBP run (no pruning); Tier-1 & 2 actionable factors are ranked and shown, Tier-3 structural traits stay in the model but are hidden (a town can't change who it is)
- Saugus analyzed as the prediction task; importance flags where it is most distinctive among actionable factors — a prediction-sharpening measure, not a causal driver
- Leave-one-out LOO r validates the predictions across all MA districts

Each model runs one RBP over 18 factors (10 of them Tier-3 structural, kept in the model but hidden from the action tables). The demographically-comparable peer towns in later exhibits come from a separate structural-only Mahalanobis match.

Model Summary (all candidates used — no pruning)			
MCAS Grades 3-8	MCAS Grade 10 (ELA)	Education Budget Share	Dropout Rate
Predicted on 18 factors	Predicted on 18 factors	Predicted on 18 factors	Predicted on 18 factors
LOO r = 0.91 · R² = 0.80	LOO r = 0.90 · R² = 0.77	LOO r = 0.79 · R² = 0.61	LOO r = 0.83 · R² = 0.62
Typical error ± 5.4 pp	Typical error ± 6.6 pp	Typical error ± 4.1 pp	Typical error ± 0.7 pp

LOO r = leave-one-out correlation of predicted vs. actual (range -1 to +1; noise below ≈ 0.15 at this sample size). R² = 1 - SSE/SST, the out-of-sample share of town-to-town variance explained (≤ r² because RBP's weighted averages compress toward the mean). Typical error = mean absolute error, in percentage points.

How to Read This Report — Three Tiers of Factors

This report matches Saugus to demographically-similar towns using Tier 3, then ranks the Tier 1 & 2 factors Saugus can actually change. Structural traits are the backdrop for finding peers — never a recommendation.

TIER 1 — Directly votable (Town Meeting / ballot)

What the town changes by a vote.

- Education's share of the municipal budget
- Residential tax rate · Proposition 2½ override · debt exclusion
- Spending above the state-required school minimum (Ch. 70)
- Reserves (free cash, stabilization) · OPEB funding · capital projects

TIER 2 — Policy / management (administration decides)

What the school department & town administration run day-to-day (funded by Tier 1).

- Chronic absenteeism — attendance & re-engagement programs
- Teacher staffing levels and pay
- How the school dollar splits between classroom and overhead
- Above-minimum school spending funded from reserves

TIER 3 — Structural (NOT changeable by vote)

What the community IS — used ONLY to find Saugus's true peer towns.

- Household income · property wealth · adult education
- Poverty · English-learner · special-education shares
- Housing tenure · district enrollment · regional economy
- → Defines the peer group; never shown as an actionable factor here.

Factor Definitions — What Each Lever Is and How It's Built

The 9 Tier-1/2 factors this report ranks. 2 are raw columns; 7 are calculated ratios. All source data is public: DESE (education feeds) and DLS Schedule A (municipal finance).

TIER 1 — Directly votable (Town Meeting / ballot)

ed_budget_share

= $\text{municipal education} \div \text{total municipal spending} \times 100$ · DLS Schedule A

→ *Share of the whole TOWN budget voted to schools — the core allocation choice.*

spend_vs_required

= $\text{in-district spending/pupil} \div \text{Chapter-70 required NSS/pupil}$ · DESE per-pupil + Chapter 70

→ *Spending above the state-required minimum — the town's discretionary 'fund-more' effort.*

fixed_costs_pct

= $\text{municipal fixed costs} \div \text{total municipal spending} \times 100$ · DLS Schedule A

→ *Budget locked in pensions/benefits/debt — the crowd-out that squeezes schools.*

TIER 2 — Policy / management (administration decides)

chronic_absenteeism_pct

= raw column (no calculation) · DESE attendance

→ *Student engagement — % of students missing $\geq 10\%$ of school days.*

avg_teacher_salary

= raw column (no calculation) · DESE staffing

→ *Teacher pay LEVEL — competitiveness for talent.*

instructional_share

= $(\text{Teachers} + \text{Other Teaching} + \text{Materials} + \text{Instr. Leadership}) \div \text{Total In-District}$ · DESE per-pupil

→ *How much of the school dollar reaches the CLASSROOM vs. overhead.*

teacher_pay_share

= $\text{Teachers (per-pupil)} \div \text{Total In-District (per-pupil)}$ · DESE per-pupil

→ *The slice of the school dollar going specifically to teacher pay.*

teachers_per_100_students

= $\text{teacher FTE} \div \text{enrollment} \times 100$ · DESE staffing + enrollment

→ *Staffing density / class size — teachers per 100 students.*

teachers_per_lowincome

= $\text{teachers-per-100-students} \div \text{low-income \%}$ · DESE staffing + enrollment + demographics

→ *Staffing RELATIVE TO NEED — teachers per unit of low-income enrollment.*

How RBP Compares Saugus to Towns Like It — Without Shrinking the Data

It focuses on towns similar to Saugus using the FULL set of MA districts, never a hand-picked subset — so the comparison keeps its statistical power.

1. Full sample, always — no pre-filtering

Every prediction uses all ~170–220 MA districts.

The full sample is what learns which towns are relevant to Saugus (the factor covariance Ω). Hand-filtering to a small peer set first would make that geometry unestimable and turn the importance scores into noise.

2. Signed weighting — contrasts, not exclusion

Each town's weight may be positive OR negative, and the weights sum to 1.

Most-relevant towns get large positive weights; less-relevant towns get small, often negative weights — used as contrasts (subtracted), as in regression. Only ~6–15 of ~170–220 towns sit near zero, and none is excluded.

3. A small effective core, plus a censored cross-check

About half the towns carry small negative weights; ~10–25 carry most of the weight.

RBP also re-predicts from only the top 80% / 50% / 20% most-relevant towns and blends those tight-peer estimates by how reliably each predicts (its fit) — the method decides how much to trust the tight group; no human picks the cutoff.

The math behind it

$$\hat{y}_{Saugus} = \sum_i w_i y_i, \quad \sum_i w_i = 1$$

$$w_i = \frac{1}{N} + \frac{\lambda^2}{n-1} (\delta_i r_i - \phi \bar{r})$$

$$r_i = -\frac{1}{2} (x_i - x_S)^T \Omega^{-1} (x_i - x_S) + \text{info}$$

$$N_{\text{eff}} = 1 / \sum_i w_i^2$$

$$w = x_S^T (X^T X)^{-1} X^T$$

The prediction is a signed weighted sum of town outcomes; the weights sum to 1.

Each weight = a uniform 1/N baseline + a tilt that turns NEGATIVE below average relevance.

Relevance r = closeness to Saugus in factor space (Mahalanobis); Ω estimated from all N towns.

Effective # of towns carrying the weight ≈ 10 –25 of ~170–220 — a focus, not a deletion.

With all factors and no censoring this is exactly OLS — signed weights are inherent to regression.

Because the relevance geometry Ω is learned from the full sample and the weights sum to 1, concentrating the prediction on a relevant core costs no stability — the instability a small hand-picked subset would create.

Combined Model Summary: All Four Outcomes

RBP factor selection results across MCAS grades 3–8, Dropout Rate, MCAS grade 10 ELA, and Education Budget Share

Model	Predicted	Actual	Gap	Direction	LOO r	Factors
MCAS Grades 3–8	25.3%	29.0%	+3.7pp	over	+0.915	18
MCAS Grade 10 (ELA)	39.2%	47.0%	+7.8pp	over	+0.896	18
Education Budget Share	42.3%	32.3%	-10.0pp	under	+0.786	18
Dropout Rate	3.8%	2.9%	-0.9pp	over	+0.827	18

Actionable Factor Cross-Reference — Which Factors Matter Across Outcomes

Actionable factors only (structural traits are matching-only). A factor that matters in multiple outcomes is a robust target.

Factor	# Models	MCAS 3–8	MCAS 10	Ed Budget	Dropout
Students chronically absent (10%+)	4	+0.389	+0.357	-0.051	+0.764
Teacher share of school spending	4	+0.321	+0.204	+0.058	+0.262
Average teacher salary	4	+0.243	+0.165	+0.056	+0.212
Spending vs Ch70 required minimum	4	+0.285	+0.171	-0.002	+0.187
Share of school \$ reaching the classroom	4	+0.252	+0.134	+0.108	+0.140
Teachers per low-income student	4	+0.244	+0.132	+0.047	+0.191
Teachers per 100 students	4	+0.207	+0.139	-0.029	+0.233
Education's share of the municipal budget	3	-0.240	-0.149	—	-0.380
Fixed costs (mostly health ins.)	1	—	—	+0.240	—

Gap = actual – predicted; the Direction column resolves the sign (for dropout, lower-than-predicted is an over-performance). Green rows = factor matters in 3+ outcomes. Cell colour: green = positive importance, red = NEGATIVE importance — the factor adds noise rather than signal for that outcome (Kritzman fn. 12; expected, not an error) — white = not a candidate in that model (—).

What This Means: MCAS Grades 3-8

Predicted 25.3pp · Actual 29.0pp · Gap +3.7pp (higher is better) · Saugus ranks 70 of 219 MA districts on this measure

The Bottom Line

Saugus's MCAS grades 3–8 proficiency (ELA + Math, meeting/exceeding) is 29.0%, 3.7pp better than its demographic prediction of 25.3% — ranking 70 of 219 MA districts, in the top half. A district with Saugus's income, poverty, and enrollment profile is predicted to land near this level.

Actionable Factors Where Saugus Stands Out

Ranked by RBP importance — where Saugus is most distinctive, Saugus vs. the statewide median.

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.39
Teacher share of school spending	0.349	0.368	+0.32
Spending vs Ch70 required minimum	1.17	1.46	+0.29
Share of school \$ reaching the classroom	51.5%	56.0%	+0.25
Teachers per low-income student	0.126	0.277	+0.24

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad, and not how much changing a factor would move the outcome. For where to act, read the peer comparison at right: the factors on which Saugus differs most from comparable towns that did better.

What Comparable Better Towns Do Differently

Same factors, for the 17 towns the model predicted like Saugus that scored better:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	21.6%	-9.6pp
Teacher share of school spending	0.349	0.360	+0.010
Spending vs Ch70 required minimum	1.17	1.33	+0.162
Share of school \$ reaching the classroom	51.5%	56.2%	+4.7pp
Teachers per low-income student	0.126	0.175	+0.049

Difference = comparable-town median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

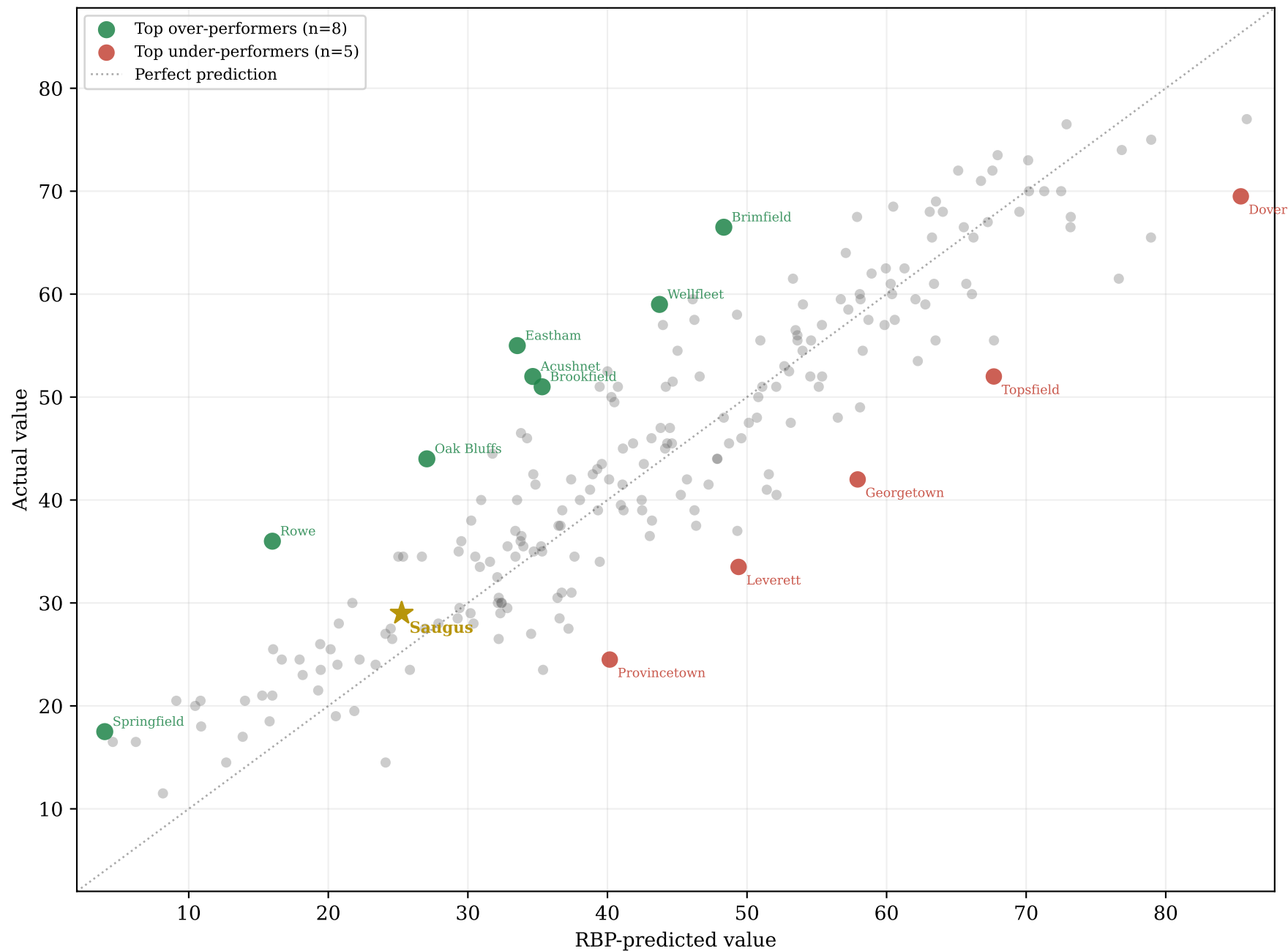
Saugus does better here than its demographic profile predicts. The factor on which Saugus is most demographically unusual is students chronically absent (10%+) (31.2% vs the 15.8% state median; peer gap -9.6pp). It both tops the RBP importance and shows the widest peer gap — two views of the same outlier status, not independent confirmation. (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) The factors below are where Saugus differs most from the towns that beat their own predictions — the most plausible places to focus attention, though the comparison is associational, not proof of cause.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns the model predicted like Saugus that scored better. Structural traits are used for peer-matching only.

Over- and Under-Performers: MCAS Grades 3-8

Saugus gap: +3.7pp · Gold star = Saugus · Green = statewide over-performers · Red = statewide under-performers

Actual vs Predicted — MCAS Grades 3-8 (LOO r = 0.915)



All MA districts shown; the biggest statewide over-/under-performers are labeled. Residual = actual – predicted. Positive = performing better than demographics suggest.

What Comparable Better Towns Do Differently: MCAS Grades 3-8

Towns the model predicted like Saugus that scored better — their actionable-factor values vs Saugus, ordered by where Saugus is furthest from these peers (gap measured in standard deviations, so different-unit factors compare)

Saugus (highlighted) vs towns predicted like Saugus that scored better
Ordered by largest peer gap (in SDs), top 8 shown

Factor	Saugus	Malden	Leominster	Peer med	Imp.
Teachers per 100 students	5.87	6.61	6.89	8.09	+0.207
Students chronically absent (10%+)	31.2	25.9	22.5	21.6	+0.389
Education's share of the municipal budget	32.3	43.1	51.8	40.9	-0.240
Share of school \$ reaching the classroom	0.51	0.60	0.57	0.56	+0.252
Spending vs Ch70 required minimum	1.17	1.20	1.14	1.33	+0.285
Average teacher salary	89,778	106,848	88,631	86,417	+0.243
Teacher share of school spending	0.35	0.39	0.34	0.36	+0.321
Teachers per low-income student	0.13	0.11	0.13	0.17	+0.244

Predicted like Saugus — but scored better

District	Actual	Predicted	vs Expected
>> Saugus	29.0	25.3	+3.7pp
Malden	34.5	25.4	+9.1pp
Leominster	34.5	25.0	+9.5pp
West Springfield	34.5	26.7	+7.8pp
Oak Bluffs	44.0	27.1	+16.9pp
Clinton	30.0	21.7	+8.3pp
Easthampton	35.0	29.3	+5.7pp
Monson	29.5	29.4	+0.1pp
Holbrook	36.0	29.5	+6.5pp
Rockland	38.0	30.2	+7.8pp
Westfield	34.5	30.5	+4.0pp
Agawam	33.5	30.9	+2.6pp
Petersham	40.0	31.0	+9.0pp

(closest 12 shown; + 5 more comparable towns within the model's prediction band)

Peer med = median across ALL comparable better-performing towns (not just the examples shown). vs Expected = how far actual beats the demographic prediction, signed so + always means better than expected (for dropout, lower-than-predicted shows as a positive 'better'). Factor values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: MCAS Grade 10 (ELA)

Predicted 39.2pp · Actual 47.0pp · Gap +7.8pp (higher is better) · Saugus ranks 39 of 168 MA districts on this measure

The Bottom Line

Saugus's grade 10 ELA proficiency is 47.0%, 7.8pp better than its demographic prediction of 39.2% — ranking 39 of 168 MA districts, in the top 25%. Note: the grade 3–8 academic base is an outcome, not an actionable factor, so it is excluded from this model — part of the over-performance shown here reflects that omitted prior strength rather than a grade-10 effect.

Actionable Factors Where Saugus Stands Out

Ranked by RBP importance — where Saugus is most distinctive, Saugus vs. the statewide median.

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.36
Teacher share of school spending	0.349	0.368	+0.20
Spending vs Ch70 required minimum	1.17	1.46	+0.17
Average teacher salary	\$89,778	\$90,677	+0.17
Education's share of the municipal budget	32.3%	48.2%	-0.15

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad, and not how much changing a factor would move the outcome. For where to act, read the peer comparison at right: the factors on which Saugus differs most from comparable towns that did better.

What Comparable Better Towns Do Differently

Same factors, for the 7 towns the model predicted like Saugus that scored better:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	20.3%	-10.9pp
Teacher share of school spending	0.349	0.363	+0.014
Spending vs Ch70 required minimum	1.17	1.34	+0.176
Average teacher salary	\$89,778	\$94,955	+5,177
Education's share of the municipal bud	32.3%	43.4%	+11.0pp

Difference = comparable-town median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

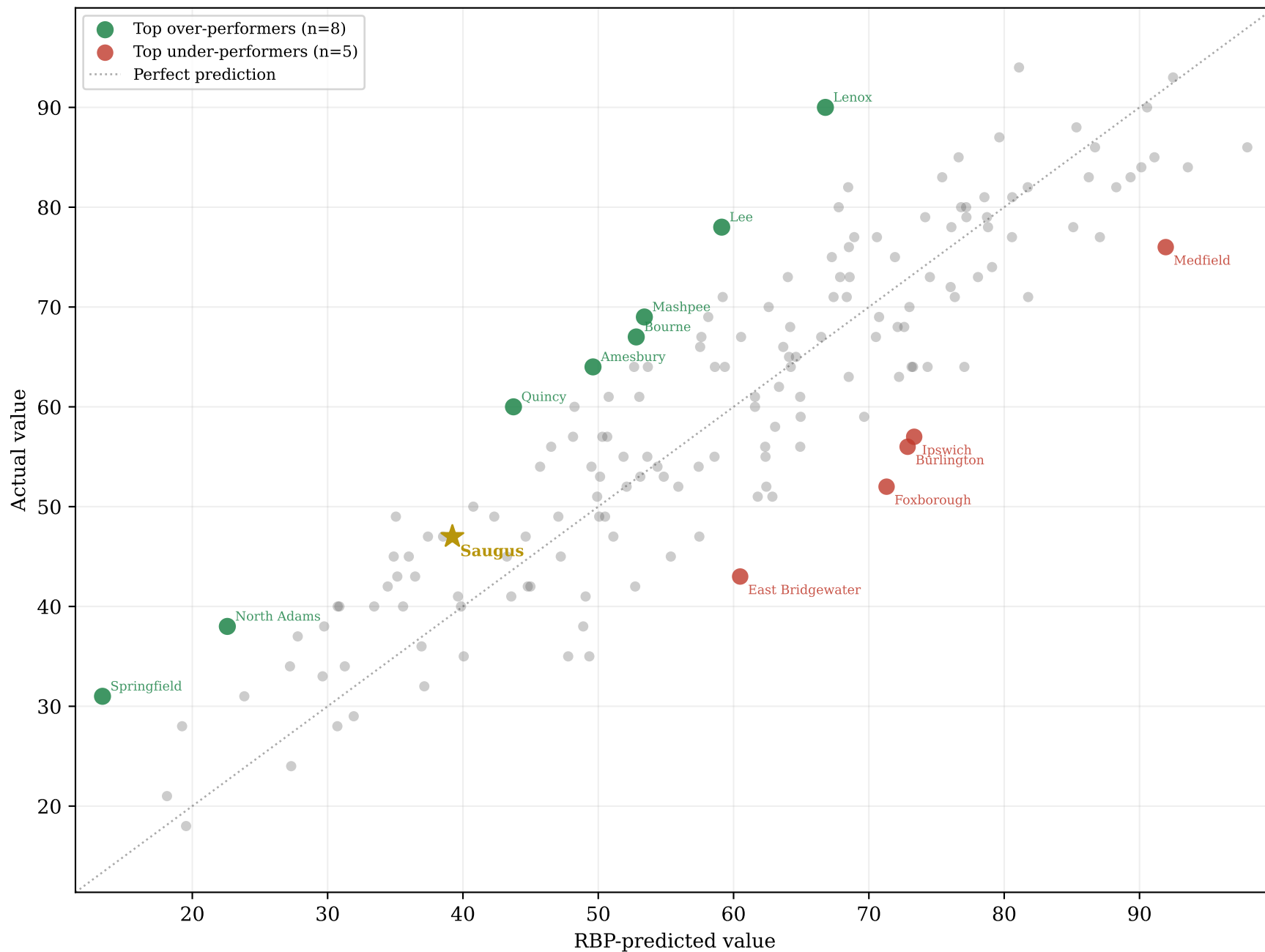
Saugus does better here than its demographic profile predicts. The strongest actionable factor is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the comparable towns that scored better is average teacher salary (+5,177, peer median – Saugus). (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) Among the actionable factors, chronic absenteeism is the clearest, with teacher pay next — Saugus clears this bar even while trailing comparable towns on attendance.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns the model predicted like Saugus that scored better. Structural traits are used for peer-matching only.

Over- and Under-Performers: MCAS Grade 10 (ELA)

Saugus gap: +7.8pp · Gold star = Saugus · Green = statewide over-performers · Red = statewide under-performers

Actual vs Predicted — MCAS Grade 10 (ELA) (LOO r = 0.896)



All MA districts shown; the biggest statewide over-/under-performers are labeled. Residual = actual – predicted. Positive = performing better than demographics suggest.

What Comparable Better Towns Do Differently: MCAS Grade 10 (ELA)

Towns the model predicted like Saugus that scored better — their actionable-factor values vs Saugus, ordered by where Saugus is furthest from these peers (gap measured in standard deviations, so different-unit factors compare)

Saugus (highlighted) vs towns predicted like Saugus that scored better
Ordered by largest peer gap (in SDs), top 8 shown

Factor	Saugus	Salem	Malden	Peer med	Imp.
Education's share of the municipal budget	32.3	43.4	43.1	43.4	-0.149
Students chronically absent (10%+)	31.2	20.3	25.9	20.3	+0.357
Teachers per 100 students	5.87	7.97	6.61	7.53	+0.139
Share of school \$ reaching the classroom	0.51	0.55	0.60	0.55	+0.134
Spending vs Ch70 required minimum	1.17	1.55	1.20	1.34	+0.171
Average teacher salary	89,778	96,388	106,848	94,955	+0.165
Teacher share of school spending	0.35	0.34	0.39	0.36	+0.204
Teachers per low-income student	0.13	0.13	0.11	0.16	+0.132

Predicted like Saugus — but scored better

District	Actual	Predicted	vs Expected
>> Saugus	47.0	39.2	+7.8pp
Salem	50.0	40.8	+9.2pp
Malden	49.0	42.3	+6.7pp
Revere	49.0	35.0	+14.0pp
Quincy	60.0	43.7	+16.3pp
Agawam	54.0	45.7	+8.3pp
Rockland	56.0	46.5	+9.5pp
Leicester	49.0	47.1	+1.9pp

Peer med = median across ALL comparable better-performing towns (not just the examples shown). vs Expected = how far actual beats the demographic prediction, signed so + always means better than expected (for dropout, lower-than-predicted shows as a positive 'better'). Factor values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Education Budget Share

Predicted 42.3pp · Actual 32.3pp · Gap -10.0pp (higher is better) · Saugus ranks 213 of 219 MA districts on this measure

The Bottom Line

Saugus's education budget share is 32.3%, 10.0pp worse than its demographic prediction of 42.3% — ranking 213 of 219 MA districts, in the bottom 10%. Unlike the academic measures, this gap reflects a municipal budgeting choice, not a demographic constraint. Note that the budget shares are compositional — they sum to the total budget — so a high importance for a competing category is partly mechanical (a larger slice elsewhere necessarily leaves a smaller one for schools), not an independent effect.

Actionable Factors Where Saugus Stands Out

Ranked by RBP importance — where Saugus is most distinctive, Saugus vs. the statewide median.

Factor	Saugus	MA median	Importance
Fixed costs (mostly health ins.)	25.3%	17.5%	+0.24
Share of school \$ reaching the classroom	51.5%	56.0%	+0.11
Teacher share of school spending	0.349	0.368	+0.06
Average teacher salary	\$89,778	\$90,677	+0.06
Students chronically absent (10%+)	31.2%	15.8%	-0.05

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad, and not how much changing a factor would move the outcome. For where to act, read the peer comparison at right: the factors on which Saugus differs most from comparable towns that did better.

What Comparable Better Towns Do Differently

Same factors, for the 100 towns the model predicted like Saugus that scored better:

Factor	Saugus	Peer median	Difference
Fixed costs (mostly health ins.)	25.3%	19.5%	-5.8pp
Share of school \$ reaching the classroom	51.5%	55.7%	+4.2pp
Teacher share of school spending	0.349	0.366	+0.016
Average teacher salary	\$89,778	\$91,460	+1,682
Students chronically absent (10%+)	31.2%	17.2%	-13.9pp

Difference = comparable-town median – Saugus. Sign direction depends on the factor; see takeaway below.

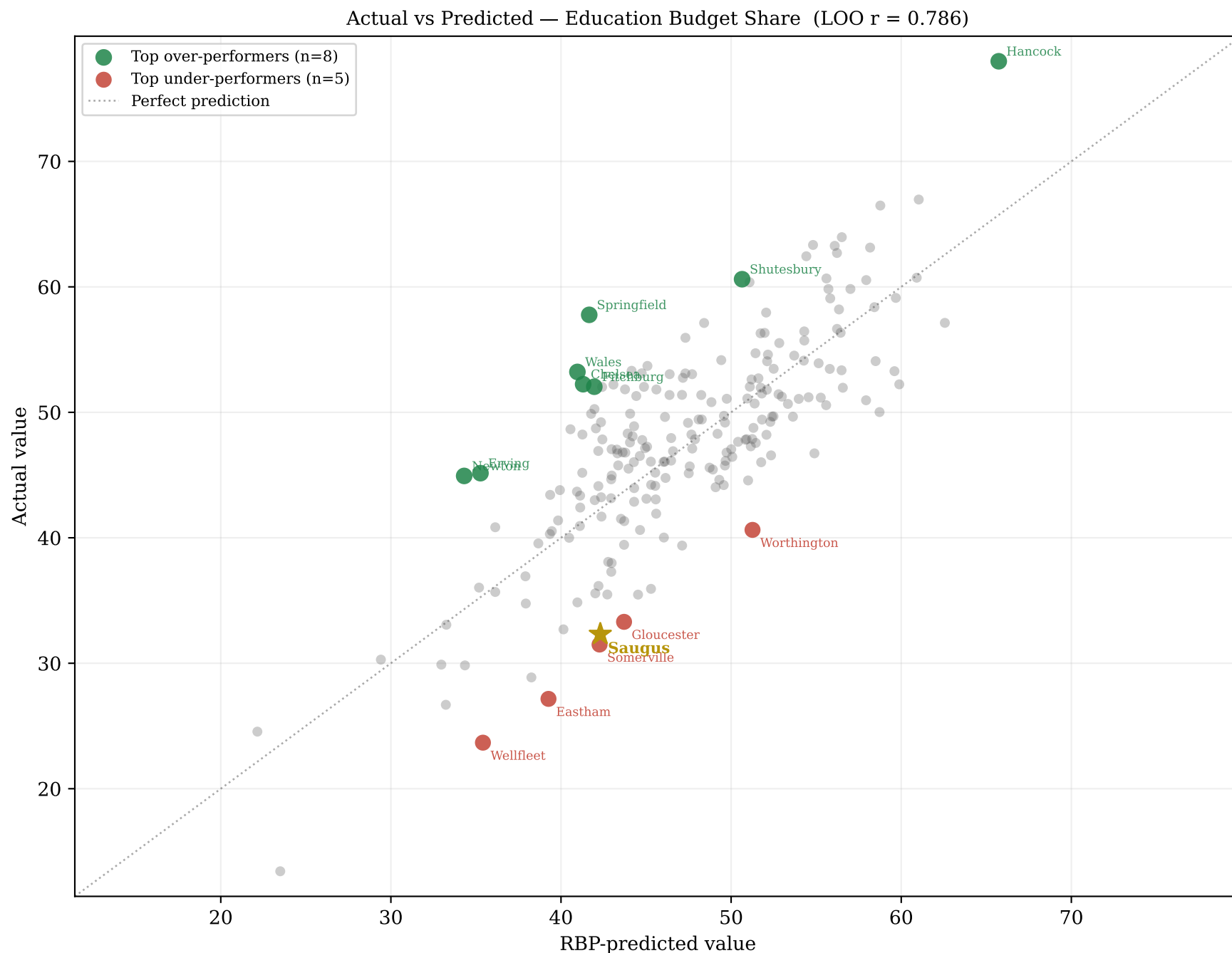
The Takeaway

Saugus does worse here than its demographic profile predicts. The strongest actionable factor is fixed costs (mostly health ins.) (25.3% vs the 17.5% state median); the widest gap to the comparable towns that scored better is average teacher salary (+1,682, peer median – Saugus). The top factors are competing budget categories that crowd out the education share — the choice here sits with Town Meeting and municipal budgeting, not the school district.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns the model predicted like Saugus that scored better. Structural traits are used for peer-matching only.

Over- and Under-Performers: Education Budget Share

Saugus gap: -10.0pp · Gold star = Saugus · Green = statewide over-performers · Red = statewide under-performers



All MA districts shown; the biggest statewide over-/under-performers are labeled. Residual = actual – predicted. Positive = performing better than demographics suggest.

What Comparable Better Towns Do Differently: Education Budget Share

Towns the model predicted like Saugus that scored better — their actionable-factor values vs Saugus, ordered by where Saugus is furthest from these peers (gap measured in standard deviations, so different-unit factors compare)

Saugus (highlighted) vs towns predicted like Saugus that scored better
Ordered by largest peer gap (in SDs), top 8 shown

Factor	Saugus	Mattapoise..	Sandwich	Peer med	Imp.
Students chronically absent (10%+)	31.2	10.2	17.9	17.2	-0.051
Fixed costs (mostly health ins.)	25.3	13.7	15.0	19.5	+0.240
Teachers per 100 students	5.87	9.22	8.07	7.97	-0.029
Share of school \$ reaching the classroom	0.51	0.56	0.49	0.56	+0.108
Spending vs Ch70 required minimum	1.17	2.09	1.90	1.43	-0.002
Teacher share of school spending	0.35	0.39	0.30	0.37	+0.058
Teachers per low-income student	0.13	0.42	0.33	0.21	+0.047
Average teacher salary	89,778	106,358	87,840	91,460	+0.056

Predicted like Saugus — but scored better

District	Actual	Predicted	vs Expected
>> Saugus	32.3	42.3	-10.0pp
Mattapoissett	49.2	42.4	+6.9pp
Sandwich	43.2	42.4	+0.9pp
Northampton	41.7	42.4	-0.7pp
Falmouth	36.2	42.2	-6.0pp
Framingham	52.0	42.4	+9.6pp
Danvers	44.1	42.2	+1.9pp
Beverly	46.9	42.2	+4.7pp
North Brookfield	47.8	42.4	+5.4pp
Wilmington	48.7	42.1	+6.7pp
Bourne	35.6	42.0	-6.4pp
Marlborough	43.0	42.0	+1.0pp
Webster	50.3	42.0	+8.3pp

(closest 12 shown; + 88 more comparable towns within the model's prediction band)

Peer med = median across ALL comparable better-performing towns (not just the examples shown). vs Expected = how far actual beats the demographic prediction, signed so + always means better than expected (for dropout, lower-than-predicted shows as a positive 'better'). Factor values shown in raw units; importance scores from RBP Exhibit 5.

What This Means: Dropout Rate

Predicted 3.8pp · Actual 2.9pp · Gap -0.9pp (lower is better) · Saugus ranks 32 of 168 MA districts on this measure

The Bottom Line

Saugus's dropout rate is 2.9%, 0.9pp better than its demographic prediction of 3.8% — ranking 32 of 168 MA districts, in the top 25%. Dropout reflects re-engagement and attendance recovery as much as demographics.

Actionable Factors Where Saugus Stands Out

Ranked by RBP importance — where Saugus is most distinctive, Saugus vs. the statewide median.

Factor	Saugus	MA median	Importance
Students chronically absent (10%+)	31.2%	15.8%	+0.76
Education's share of the municipal budget	32.3%	48.2%	-0.38
Teacher share of school spending	0.349	0.368	+0.26
Teachers per 100 students	5.87	8.07	+0.23
Average teacher salary	\$89,778	\$90,677	+0.21

Importance measures how much each factor sharpens the search for Saugus's true comparison districts — not whether higher values are good or bad, and not how much changing a factor would move the outcome. For where to act, read the peer comparison at right: the factors on which Saugus differs most from comparable towns that did better.

What Comparable Better Towns Do Differently

Same factors, for the 12 towns the model predicted like Saugus that scored better:

Factor	Saugus	Peer median	Difference
Students chronically absent (10%+)	31.2%	23.4%	-7.8pp
Education's share of the municipal bud	32.3%	46.9%	+14.6pp
Teacher share of school spending	0.349	0.354	+0.005
Teachers per 100 students	5.87	7.52	+1.65
Average teacher salary	\$89,778	\$89,740	-38

Difference = comparable-town median – Saugus. Sign direction depends on the factor; see takeaway below.

The Takeaway

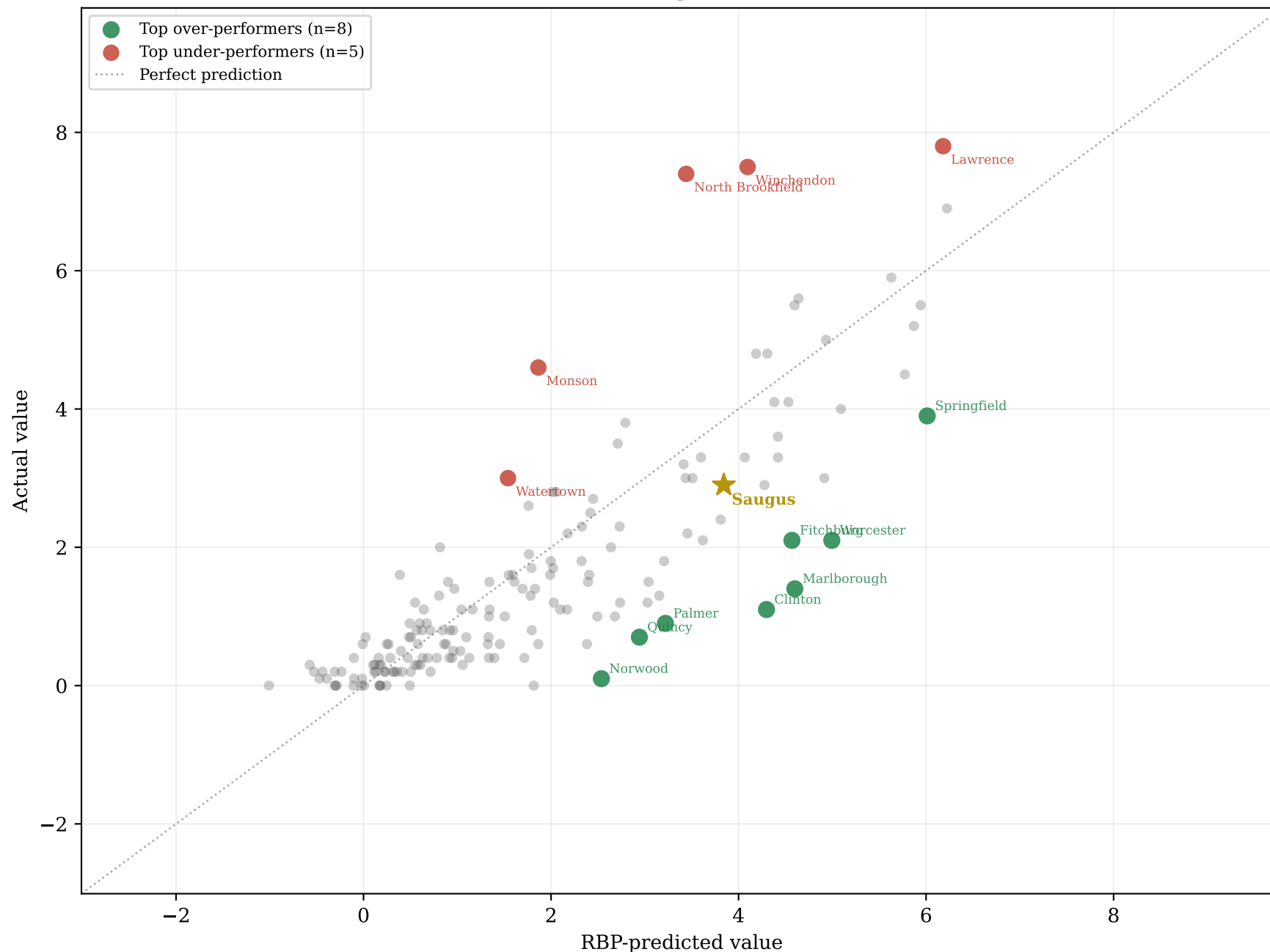
Saugus does better here than its demographic profile predicts. The strongest actionable factor is students chronically absent (10%+) (31.2% vs the 15.8% state median); the widest gap to the comparable towns that scored better is average teacher salary (-38, peer median – Saugus). (Seed-stability: the top factor is seed-stable; the leading factors below it recur across random grids but their order and importance values jump run-to-run — read them as a co-equal group, not a strict ranking.) Chronic absenteeism is the factor to watch — the main risk to this result if it climbs.

Left: top actionable factors by RBP importance, Saugus vs. the state median. Right: the same factors for the towns the model predicted like Saugus that scored better. Structural traits are used for peer-matching only.

Over- and Under-Performers: Dropout Rate

Saugus gap: -0.9pp · Gold star = Saugus · Green = statewide over-performers · Red = statewide under-performers

Actual vs Predicted — Dropout Rate (LOO r = 0.827)



All MA districts shown; the biggest statewide over-/under-performers are labeled. Residual = actual – predicted. Positive = performing better than demographics suggest.

What Comparable Better Towns Do Differently: Dropout Rate

Towns the model predicted like Saugus that scored better — their actionable-factor values vs Saugus, ordered by where Saugus is furthest from these peers (gap measured in standard deviations, so different-unit factors compare)

Saugus (highlighted) vs towns predicted like Saugus that scored better
Ordered by largest peer gap (in SDs), top 8 shown

Factor	Saugus	Greenfield	Chicopee	Peer med	Imp.
Education's share of the municipal budget	32.3	34.8	49.2	46.9	-0.380
Teachers per 100 students	5.87	6.94	8.18	7.52	+0.233
Students chronically absent (10%+)	31.2	27.8	27.3	23.4	+0.764
Share of school \$ reaching the classroom	0.51	0.48	0.57	0.54	+0.140
Spending vs Ch70 required minimum	1.17	1.59	1.34	1.29	+0.187
Teacher share of school spending	0.35	0.28	0.36	0.35	+0.262
Teachers per low-income student	0.13	0.11	0.12	0.13	+0.191
Average teacher salary	89,778	73,650	90,848	89,740	+0.212

Predicted like Saugus — but scored better

District	Actual	Predicted	vs Expected
>> Saugus	2.9	3.8	+0.9pp better
Greenfield	2.4	3.8	+1.4pp better
Chicopee	2.1	3.6	+1.5pp better
Taunton	2.2	3.5	+1.3pp better
Clinton	1.1	4.3	+3.2pp better
Palmer	0.9	3.2	+2.3pp better
Salem	1.8	3.2	+1.4pp better
Agawam	1.3	3.2	+1.9pp better
Fitchburg	2.1	4.6	+2.5pp better
Marlborough	1.4	4.6	+3.2pp better
Somerville	1.5	3.0	+1.5pp better
Leominster	1.2	3.0	+1.8pp better
Quincy	0.7	2.9	+2.2pp better

Peer med = median across ALL comparable better-performing towns (not just the examples shown). vs Expected = how far actual beats the demographic prediction, signed so + always means better than expected (for dropout, lower-than-predicted shows as a positive 'better'). Factor values shown in raw units; importance scores from RBP Exhibit 5.

Optimum Profile — What Overachievers Look Like

Each grid shows actionable factor values for Saugus vs over-performer towns. Showing both peer groups tests whether the finding holds across definitions.

Overachievers — 40 MA districts that most consistently beat their demographic prediction

Feature	Saugus	Springfield	Lenox	Pelham	Falmouth	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	32.4%	18.3%	10.0%	24.0%	17.9%	-13.2pp	40
Teachers / 100 students	5.87	7.37	10.1	9.41	9.14	8.08	+2.21	40
Teachers per low-income student	0.126	0.087	0.414	0.357	0.231	0.209	+0.083	40
Spending vs Ch70 minimum (×)	1.17	1.43	1.73	1.65	1.75	1.49	+0.318	40
School spending vs. property wealth	0.082	0.363	0.082	0.144	0.054	0.116	+0.035	40
Teacher share of school \$	0.349	0.291	0.369	0.309	0.360	0.363	+0.014	40
Shown: Springfield, Lenox, Pelham, Falmouth (+ 36 more in median)								

Demographically Similar Overachievers — the 12 within 2σ Mahalanobis distance of Saugus

Feature	Saugus	Winthrop	Northbridge	West Boyls...	Mashpee	Peer med	Gap	N
Chronic absenteeism (%)	31.2%	17.4%	20.5%	15.6%	18.0%	17.9%	-13.2pp	12
Teachers / 100 students	5.87	7.78	7.08	8.09	8.42	8.05	+2.18	12
Teachers per low-income student	0.126	0.209	0.184	0.257	0.206	0.209	+0.083	12
Spending vs Ch70 minimum (×)	1.17	1.25	1.18	1.40	1.62	1.45	+0.286	12
School spending vs. property wealth	0.082	0.107	0.136	0.129	0.055	0.107	+0.025	12
Teacher share of school \$	0.349	0.361	0.390	0.399	0.336	0.367	+0.018	12
Shown: Winthrop, Northbridge, West Boylston, Mashpee (+ 8 more in median)								

Similarity metric: standardized Mahalanobis distance $d_M = \sqrt{(\Delta z \cdot R^{-1} \cdot \Delta z^*)}$, z = population z-score and R = correlation matrix of all MA districts on (median_hh_income, equalized_income, low_income_pct, total_enrollment, pct_bachelors_plus, ell_pct). Threshold: $d_M \leq 2$ (2σ). Pre-specified; not tuned to result. Gap = peer median – Saugus. Red = Saugus below peer median, Green = at or above.